

## Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11  
priors compared

32,641  
held-out plate appearances

3.05%  
held-out HR rate

0.5999  
best AUC · KNN comps

served prior

### Held-out scores

Fitted on plate appearances before 2026-08-27, scored on the 32,641 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

| PRIOR                                                                                          | AUC    | TOP-DECILE LIFT | BRIER   | LOG LOSS | CAL. RMSE |
|------------------------------------------------------------------------------------------------|--------|-----------------|---------|----------|-----------|
| <b>Flat league</b><br>League HR rate as the prior for every hitter.                            | 0.5836 | 1.55×           | 0.02952 | 0.13556  | 0.00462   |
| <b>KNN comps</b><br>Prior is the pooled rate of the nearest comparables.                       | 0.5999 | 1.52×           | 0.02949 | 0.13463  | 0.00494   |
| <b>SVR</b><br>Support-vector regression on the same feature matrix.                            | 0.5969 | 1.48×           | 0.02953 | 0.13522  | 0.00618   |
| <b>Random forest</b><br>Forest regression; a learned similarity metric.                        | 0.5958 | 1.47×           | 0.02952 | 0.13504  | 0.00551   |
| <b>Core (KNN+SVR)</b><br>Blend of KNN and SVR, no forest.                                      | 0.5999 | 1.52×           | 0.02949 | 0.13463  | 0.00494   |
| <b>Forest blend</b><br>KNN + SVR + forest, weights fitted out-of-fold.                         | 0.5999 | 1.52×           | 0.02949 | 0.13463  | 0.00494   |
| <b>Linear regression</b><br>Ordinary least squares on the same features.                       | 0.5965 | 1.47×           | 0.02951 | 0.13498  | 0.00436   |
| <b>Logistic regression</b><br>Binomial logistic on the per-PA home run outcome.                | 0.5976 | 1.54×           | 0.02953 | 0.13512  | 0.00680   |
| <b>Neural net (torch)</b><br>Two-layer PyTorch MLP, PA-weighted, trained on the same features. | 0.5964 | 1.46×           | 0.02951 | 0.13496  | 0.00557   |
| <b>XGBoost</b><br>Gradient-boosted trees, depth 3 with subsampling and L2.                     | 0.5971 | 1.45×           | 0.02953 | 0.13503  | 0.00625   |

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 248 knn 248 svr 248 rf 248 core 248 ensemble 248  
linear 248 logistic 248 nn 248 xgb 248

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,547 PA, actual HR rate 2.34%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

| PRIOR                                                                                          | AUC    | TOP-DECILE LIFT | BRIER   | LOG LOSS | CAL. RMSE |
|------------------------------------------------------------------------------------------------|--------|-----------------|---------|----------|-----------|
| <b>Flat league</b><br>League HR rate as the prior for every hitter.                            | 0.5526 | 1.44×           | 0.02288 | 0.11153  | 0.00903   |
| <b>KNN comps</b><br>Prior is the pooled rate of the nearest comparables.                       | 0.5744 | 1.20×           | 0.02296 | 0.11191  | 0.01517   |
| <b>SVR</b><br>Support-vector regression on the same feature matrix.                            | 0.5731 | 1.32×           | 0.02313 | 0.11507  | 0.01888   |
| <b>Random forest</b><br>Forest regression; a learned similarity metric.                        | 0.5821 | 1.08×           | 0.02314 | 0.11331  | 0.01866   |
| <b>Core (KNN+SVR)</b><br>Blend of KNN and SVR, no forest.                                      | 0.5744 | 1.20×           | 0.02296 | 0.11191  | 0.01517   |
| <b>Forest blend</b><br>KNN + SVR + forest, weights fitted out-of-fold.                         | 0.5744 | 1.20×           | 0.02296 | 0.11191  | 0.01517   |
| <b>Linear regression</b><br>Ordinary least squares on the same features.                       | 0.5768 | 0.60×           | 0.02307 | 0.11384  | 0.01888   |
| <b>Logistic regression</b><br>Binomial logistic on the per-PA home run outcome.                | 0.5769 | 0.48×           | 0.02314 | 0.11364  | 0.02055   |
| <b>Neural net (torch)</b><br>Two-layer PyTorch MLP, PA-weighted, trained on the same features. | 0.5743 | 1.32×           | 0.02303 | 0.11288  | 0.01760   |
| <b>XGBoost</b><br>Gradient-boosted trees, depth 3 with subsampling and L2.                     | 0.5886 | 1.08×           | 0.02315 | 0.11327  | 0.01986   |

Slate agreement

| PRIOR                | SHARED | ITS TOP SIX                                                                                                                         |
|----------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------|
| Served (full season) | 6/6    | Matt Olson 17.0%, Ronald Acuña Jr. 16.4%, Fernando Tatis Jr. 15.8%, Shohei Ohtani 14.9%, Drake Baldwin 14.8%, Jackson Chourio 14.5% |
| Flat league          | 6/6    | Fernando Tatis Jr. 15.7%, Ronald Acuña Jr. 15.7%, Matt Olson 15.4%, Drake Baldwin 14.5%, Jackson Chourio 14.5%, Shohei Ohtani 13.4% |
| KNN comps            | 6/6    | Matt Olson 16.6%, Ronald Acuña Jr. 16.4%, Fernando Tatis Jr. 16.1%, Shohei Ohtani 15.0%, Drake Baldwin 15.0%, Jackson Chourio 14.9% |
| SVR                  | 6/6    | Matt Olson 17.2%, Ronald Acuña Jr. 16.5%, Fernando Tatis Jr. 15.9%, Drake Baldwin 15.2%, Shohei Ohtani 15.1%, Jackson Chourio 15.0% |
| Random forest        | 6/6    | Matt Olson 17.0%, Ronald Acuña Jr. 16.3%, Fernando Tatis Jr. 15.7%, Drake Baldwin 15.6%, Shohei Ohtani 15.2%, Jackson Chourio 15.1% |
| Core (KNN+SVR)       | 6/6    | Matt Olson 16.6%, Ronald Acuña Jr. 16.4%, Fernando Tatis Jr. 16.1%, Shohei Ohtani 15.0%, Drake Baldwin 15.0%, Jackson Chourio 14.9% |
| Forest blend         | 6/6    | Matt Olson 16.6%, Ronald Acuña Jr. 16.4%, Fernando Tatis Jr. 16.1%, Shohei Ohtani 15.0%, Drake Baldwin 15.0%, Jackson Chourio 14.9% |
| Linear regression    | 6/6    | Matt Olson 16.7%, Fernando Tatis Jr. 16.3%, Ronald Acuña Jr. 16.2%, Drake Baldwin 15.3%, Shohei Ohtani 14.9%, Jackson Chourio 14.9% |
| Logistic regression  | 6/6    | Matt Olson 17.3%, Fernando Tatis Jr. 16.4%, Ronald Acuña Jr. 16.3%, Shohei Ohtani 15.3%, Drake Baldwin 15.1%, Jackson Chourio 14.9% |
| Neural net (torch)   | 6/6    | Matt Olson 17.0%, Fernando Tatis Jr. 16.4%, Ronald Acuña Jr. 16.3%, Shohei Ohtani 15.2%, Drake Baldwin 15.0%, Jackson Chourio 15.0% |
| XGBoost              | 6/6    | Matt Olson 17.0%, Ronald Acuña Jr. 16.2%, Fernando Tatis Jr. 15.7%, Drake Baldwin 15.6%, Shohei Ohtani 15.4%, Jackson Chourio 15.2% |

Hitter by hitter

The top 40 bats by the served probability.

| MLB Season Performance Analysis - Q3 2024 |                      |             |           |       |               |                |              |                   |                     |                |  |
|-------------------------------------------|----------------------|-------------|-----------|-------|---------------|----------------|--------------|-------------------|---------------------|----------------|--|
| HITTER                                    | SERVED (FULL SEASON) | FLAT LEAGUE | KNN COMPS | SVR   | RANDOM FOREST | CORE (KNN+SVR) | FOREST BLEND | LINEAR REGRESSION | LOGISTIC REGRESSION | NEURAL NET (TO |  |
| 1Matt Olson                               | slate                | 17.0%       | 15.4%     | 16.6% | 17.2%         | 17.0%          | 16.6%        | 16.6%             | 16.7%               | 17.3%          |  |
| 2Ronald Acuña Jr.                         | FL                   | 16.4%       | 15.7%     | 16.4% | 16.5%         | 16.3%          | 16.4%        | 16.4%             | 16.2%               | 16.3%          |  |
| 3Fernando Tatis Jr.                       | slate                | 15.8%       | 15.7%     | 16.1% | 15.9%         | 15.7%          | 16.1%        | 16.1%             | 16.3%               | 16.4%          |  |
| 4Shohei Ohtani                            | slate                | 14.9%       | 13.4%     | 15.0% | 15.1%         | 15.2%          | 15.0%        | 15.0%             | 14.9%               | 15.3%          |  |
| 5Drake Baldwin                            | ATL                  | 14.8%       | 14.5%     | 15.0% | 15.2%         | 15.6%          | 15.0%        | 15.0%             | 15.3%               | 15.1%          |  |
| 6Jackson Chourio                          | slate                | 14.5%       | 14.5%     | 14.9% | 15.0%         | 15.1%          | 14.9%        | 14.9%             | 14.9%               | 14.9%          |  |
| 7Jackson Merrill                          | SD                   | 13.8%       | 12.4%     | 13.7% | 13.7%         | 13.7%          | 13.7%        | 13.7%             | 13.7%               | 14.0%          |  |
| 8Teoscar Hernández                        | SD                   | 13.2%       | 12.5%     | 13.3% | 13.4%         | 13.6%          | 13.3%        | 13.3%             | 13.5%               | 13.5%          |  |
| 9Manny Machado                            | SD                   | 13.1%       | 12.7%     | 13.2% | 13.6%         | 13.3%          | 13.2%        | 13.2%             | 13.4%               | 13.5%          |  |
| 10Will Smith                              | LAD                  | 13.0%       | 12.7%     | 13.0% | 13.1%         | 13.6%          | 13.0%        | 13.0%             | 13.6%               | 13.4%          |  |
| 11William Contreras                       | AS                   | 12.5%       | 12.3%     | 12.5% | 12.3%         | 12.3%          | 12.5%        | 12.5%             | 12.4%               | 12.4%          |  |
| 12Mookie Betts                            | LAD                  | 12.1%       | 12.1%     | 11.9% | 12.6%         | 11.9%          | 11.9%        | 11.9%             | 12.3%               | 12.3%          |  |
| 13Jake Bauers                             | MIL                  | 12.0%       | 11.2%     | 12.3% | 12.2%         | 12.2%          | 12.3%        | 12.3%             | 12.1%               | 12.0%          |  |
| 14Austin Riley                            | ATL                  | 12.0%       | 11.0%     | 11.7% | 11.7%         | 11.5%          | 11.7%        | 11.7%             | 11.7%               | 11.8%          |  |
| 15Garrett Mitchell                        | IL                   | 11.6%       | 11.1%     | 11.6% | 10.2%         | 10.8%          | 11.6%        | 11.6%             | 11.0%               | 10.9%          |  |
| 16Michael Harris II                       | FL                   | 11.1%       | 11.2%     | 11.0% | 11.6%         | 11.0%          | 11.0%        | 11.0%             | 11.3%               | 11.6%          |  |
| 17Ty France                               | SD                   | 10.8%       | 10.6%     | 11.0% | 10.9%         | 11.1%          | 11.0%        | 11.0%             | 11.0%               | 11.0%          |  |
| 18Brice Turang                            | MIL                  | 10.7%       | 11.0%     | 10.5% | 10.8%         | 10.4%          | 10.5%        | 10.5%             | 10.1%               | 10.1%          |  |
| 19Andy Pages                              | LAD                  | 10.0%       | 9.5%      | 9.6%  | 9.4%          | 9.7%           | 9.6%         | 9.6%              | 9.5%                | 9.4%           |  |
| 20Sean Murphy                             | ATL                  | 9.9%        | 9.5%      | 10.1% | 9.4%          | 10.2%          | 10.1%        | 10.1%             | 10.3%               | 9.7%           |  |
| 21Mike Yastrzemski                        | KL                   | 9.7%        | 9.8%      | 9.9%  | 9.8%          | 9.7%           | 9.9%         | 9.9%              | 9.6%                | 9.5%           |  |
| 22Ozzie Albies                            | ATL                  | 9.7%        | 10.1%     | 9.6%  | 10.3%         | 10.3%          | 9.6%         | 9.6%              | 10.0%               | 9.9%           |  |
| 23Freddie Freeman                         | LAN                  | 9.6%        | 9.3%      | 9.3%  | 9.2%          | 9.1%           | 9.3%         | 9.3%              | 9.4%                | 9.2%           |  |
| 24Christian Yelich                        | MIL                  | 9.4%        | 9.6%      | 9.7%  | 9.3%          | 9.6%           | 9.7%         | 9.7%              | 9.8%                | 9.4%           |  |
| 25Xander Bogaerts                         | SD                   | 9.0%        | 9.0%      | 8.8%  | 8.6%          | 8.8%           | 8.8%         | 8.8%              | 9.0%                | 8.7%           |  |
| 26Kyle Tucker                             | LAD                  | 9.0%        | 9.1%      | 9.1%  | 8.8%          | 9.0%           | 9.1%         | 9.1%              | 8.9%                | 8.7%           |  |
| 27Luis Campusano                          | SD                   | 7.7%        | 7.7%      | 7.9%  | 7.6%          | 7.8%           | 7.9%         | 7.9%              | 7.9%                | 7.6%           |  |
| 28Joey Ortiz                              | MIL                  | 7.3%        | 8.4%      | 7.5%  | 7.6%          | 7.5%           | 7.5%         | 7.5%              | 7.4%                | 7.4%           |  |

| HITTER             | SERVED (FULL SEASON) | FLAT LEAGUE | KNN COMPS | SVR  | RANDOM FOREST | CORE (KNN+SVR) | FOREST BLEND | LINEAR REGRESSION | LOGISTIC REGRESSION | NEURAL NET (TO |
|--------------------|----------------------|-------------|-----------|------|---------------|----------------|--------------|-------------------|---------------------|----------------|
| 29Mauricio Dubón   | 7.1%                 | 8.6%        | 7.2%      | 7.6% | 7.4%          | 7.2%           | 7.2%         | 7.1%              | 7.4%                |                |
| 30Dustin Harris    | 6.8%                 | 8.2%        | 6.7%      | 5.8% | 6.0%          | 6.7%           | 6.7%         | 6.6%              | 6.4%                |                |
| 31Tommy Edman      | 6.1%                 | 6.6%        | 6.0%      | 6.1% | 5.6%          | 6.0%           | 6.0%         | 6.1%              | 6.0%                |                |
| 32Ha-Seong Kim     | 5.7%                 | 7.0%        | 5.0%      | 4.4% | 4.3%          | 5.0%           | 5.0%         | 5.4%              | 5.3%                |                |
| 33Cooper Pratt     | 5.4%                 | 7.0%        | 5.4%      | 4.9% | 5.1%          | 5.4%           | 5.4%         | 4.9%              | 4.9%                |                |
| 34Sal Frelick      | 4.9%                 | 6.0%        | 4.8%      | 4.6% | 4.6%          | 4.8%           | 4.8%         | 4.7%              | 4.7%                |                |
| 35Jake Cronenworth | 4.7%                 | 5.9%        | 4.8%      | 4.8% | 4.8%          | 4.8%           | 4.8%         | 4.8%              | 4.8%                |                |
| 36David Hamilton   | 4.3%                 | 5.7%        | 4.5%      | 4.3% | 4.0%          | 4.5%           | 4.5%         | 4.3%              | 4.3%                |                |
| 37Luis Lara        | 4.3%                 | 6.0%        | 4.6%      | 3.7% | 4.0%          | 4.6%           | 4.6%         | 3.6%              | 4.2%                |                |

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.